Topic 15: Organic Chemistry: Carbonyls, Carboxylic Acids and Chirality

15A: Chirality

Students will be assessed on their ability to:

15.1	know that optical isomerism is a result of chirality in molecules with a single chiral centre
15.2	understand that optical isomerism results from chiral centre(s) in a molecule with asymmetric carbon atom(s) and that optical isomers (enantiomers) are object and non-superimposable mirror images and be able to draw 3D diagrams of these optical isomers
15.3	know that optical activity is the ability of a single optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light in molecules containing a single chiral centre
15.4	know what is meant by the term 'racemic mixture'
15.5	be able to use data on optical activity of reactants and products as evidence for S_N1 and S_N2 mechanisms and addition to carbonyl compounds

15B: Carbonyl compounds

Students will be assessed on their ability to:

15.6	understand the nomenclature of aldehydes and ketones and be able to draw their structural, displayed and skeletal formulae
15.7	understand that aldehydes and ketones: i do not form intermolecular hydrogen bonds and this affects their physical properties ii can form hydrogen bonds with water and this affects their solubility
15.8	understand the reactions of carbonyl compounds with: i Fehling’s or Benedict’s solution, Tollens’ reagent and acidified dichromate(VI) ions *In equations, the oxidising agent can be represented as [O]*. ii lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane) *In equations, the reducing agent can be represented by [H]*. iii HCN, in the presence of KCN, as a nucleophilic addition reaction, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism iv 2,4-dinitrophenylhydrazine (2,4-DNPH), as a qualitative test for the presence of a carbonyl group and to identify a carbonyl compound given data of the melting temperatures of derivatives *The equation for this reaction is not required.* v iodine in the presence of alkali (the iodoform test)
	Further suggested practical: Reactions of aldehydes and ketones given in 15.8 i, iv and v

15C: Carboxylic acids

Students will be assessed on their ability to:

15.9	understand the nomenclature of carboxylic acids and be able to draw their structural, displayed and skeletal formulae
15.10	understand that hydrogen bonding affects the physical properties of carboxylic acids, in relation to their boiling temperatures and solubility
15.11	understand that carboxylic acids can be prepared by the oxidation of alcohols or aldehydes and the hydrolysis of nitriles
15.12	understand the reactions of carboxylic acids with: i lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane) ii bases to produce salts iii phosphorus(V) chloride (phosphorus pentachloride) iv alcohols in the presence of an acid catalyst
	Further suggested practicals: i solubility of a range of carboxylic acids, aldehydes and ketones ii preparation of carboxylic acids by the oxidation of alcohols and aldehydes iii reactions of carboxylic acids given in 15.12 ii, iii and iv

15D: Carboxylic acid derivatives

Students will be assessed on their ability to:

15.13	understand the nomenclature of acyl chlorides and esters and be able to draw their structural, displayed and skeletal formulae
15.14	understand the reactions of acyl chlorides with: i water ii alcohols iii concentrated ammonia iv amines
15.15	understand the hydrolysis reactions of esters, in acidic and alkaline solution
15.16	understand how polyesters, such as terylene, are formed by condensation polymerisation reactions.
	Further suggested practicals: i demonstration of the reactions of ethanoyl chloride given in 15.14 i, ii and iii ii the preparation of esters such as ethyl ethanoate as a solvent or a pineapple flavouring iii hydrolysis of an ester

15E: Spectroscopy and chromatography

Knowledge of the concepts introduced in Unit 2, Topic 10D: Mass spectra and IR will be assumed and extended in this topic

Students will be assessed on their ability to:

15.17	be able to use data from mass spectra to: i suggest possible structures of a simple organic compound given accurate relative molecular masses ii calculate the accurate relative molecular mass of a compound, given accurate relative atomic masses to four decimal places
15.18	understand that carbon-13, (^{13}C) NMR spectroscopy provides information about the positions of ^{13}C atoms in a molecule
15.19	be able to use data from ^{13}C NMR spectroscopy to: i predict the different environments for carbon atoms present in a molecule, given values of chemical shift, δ ii justify the number of peaks present in a ^{13}C NMR spectrum in terms of the number of carbon atoms in different environments
15.20	be able to use both low and high resolution proton NMR spectroscopy to: i predict the different types of proton present in a molecule, given values of chemical shift, δ ii relate relative peak areas, or ratio number of protons, to the relative numbers of ^{1}H atoms in different environments iii deduce the splitting patterns of adjacent, non-equivalent protons using the (n+1) rule and hence suggest the possible structures for a molecule iv predict the chemical shifts and splitting patterns of the ^{1}H atoms in a given molecule
15.21	know that chromatography separates components of a mixture using a mobile phase and a stationary phase
15.22	be able to calculate R_f values from one-way chromatograms in paper and thin-layer chromatography (TLC) and understand reasons for differences in R_f values
15.23	know that high-performance liquid chromatography, HPLC, and gas chromatography, GC, are types of column chromatography that separate substances because of different retention times in the column and may be used in conjunction with mass spectrometry, in applications such as forensics or drug testing in sport